VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



CO5151 – Advanced Agentic Artificial Intelligence

D2 Interim Report

LuatCheck: A Local, Source-Grounded Assistant over Vietnamese Regulations

Application Track – Group 8

[PENDING: Draft of 24 October 2026. Human audit of the judge, member contribution evidence, the commit hash and the Mac latency of the chosen system are still pending.]

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Repository: https://github.com/tuandung222/LuatCheck-D2

Reported revision: **[PENDING: insert the full final commit hash after freezing the source]**

Access: **[PENDING: confirm that the instructor can read the repository]**

Status: On track, with two risks: the gain of the revised system is not yet confirmed on unseen tasks, and the judge is not yet checked against human labels (Section 7).

HO CHI MINH CITY, OCTOBER 2026

Summary. LuatCheck answers a question only from a document collection, cites the sections it used, checks each citation against the source text in code, and refuses or asks for clarification when the sources do not answer. The chosen system, *LuatCheck r2*, is one bounded agent. Its controller is a 4-billion-parameter model that runs on a laptop (a lighter 2B option exists). It uses read-only search and read tools, keyword retrieval with a neural reranker, a citation verifier with bounded repair, and input guardrails.

Evaluated: two 50-task Vietnamese benchmarks (university academic regulations, labour law), three configurations (verifier off, verifier on, verifier and reranker on), three seeds, plus two diagnostic baselines; 1,090 recorded trials per system. Success is judged by an LLM, as a lower bound; the check against human labels is pending.

Result: the chosen system reaches 60.7% (HCMUT) and 56.7% (labour) strict success; the first design (v3, 2B) reached 14.0% and 18.7%. The revised design r2 was made after we had seen the v3 results on the same tasks, so these gains are optimistic; D3 confirms them on a new held-out benchmark. In the ablation, neither the verifier nor the reranker raised strict success on these benchmarks. **Not yet available, in red:** human audit of the judge, member contribution evidence, commit hash, Mac latency of the chosen system.

1 Changes from the Proposal

Scope and users. The proposal (D1, Application Track) targeted employees and HR staff asking labour-law questions. For D2 we added a second corpus, the academic-affairs regulations of HCMUT used by students and advisers, to test whether the same code serves another domain: wording, tool names and ID rules are per-domain configuration. We claim only what was evaluated: ten HCMUT documents and ten labour-law documents (the 2019 Labour Code with two laws, six decrees and one circular of 2020–2025), snapshot of 22 October 2026, later amendments unchecked. LuatCheck is not legal advice and has not been tried by real users. Intended scenarios: a student looks up a rule and opens the cited section; an HR officer checks a labour-law answer against the cited article; a careless or hostile user pastes a document with injected instructions, and the system must keep its tool boundary and refuse or clarify.

Design history. The first design (v3, controller 2B) was frozen and run first; it reached 12–19% strict success. An error analysis of its recorded runs found four causes:

- the controller's own search queries retrieved worse than the question itself (an offline projection puts first-search Recall@3 at 0.71 and 0.62, against 0.95 and 0.98 if the question were the query);
- it asked for clarification as an exit whenever evidence was missing (37% and 30% of answerable tasks without the verifier), although no task expects one;
- after a verifier rejection it refused in 56% and 53% of cases instead of repairing the answer;
- most rejections were exact-quote mismatches caused by elided quotes.

The revision r2 addresses them. The first search is anchored on the question, the clarify and refuse rules are explicit, an answer needs a prior search, and the model's reasoning text is capped. Verifier v2 accepts elided and wrapped quotes, checks numbers in the claim's own sentence and shows the closest source span in its feedback. r2 runs with a 4B controller (chosen) and a 2B controller (lighter option).

Commitments and their state. Table 1 compares the D1 commitments due at the D2 checkpoint with their state. We received no feedback on the D1 pitch (22 October), so no change is attributed to feedback.

2 Implemented Architecture

LuatCheck is one agent running a ReAct-style loop [8] over retrieval-augmented generation [6] (Figure 1). Before any model call, input guardrails (regular-expression patterns, shared and per domain; no learned classifier) refuse obviously unsafe input, and the question is wrapped as untrusted text. At each step the controller emits one JSON action: search, read, answer, clarify or refuse. A search ranks records with BM25 and TF-IDF fused by reciprocal rank fusion (RRF), rescores the top 20 with a neural cross-encoder reranker and returns the best five as the observation.

An answer goes through the L1 verifier, which checks in code that each cited record exists and was retrieved, that the quoted sentence appears in it, and that the claim's terms and numbers match the

Proposal commitment (D1)	State	Delta and reason
Budgeted loop, schema output, L1 citation check	Done	Runs end to end; extended with bounded repair and verifier v2.
Qwen3 4B/8B controllers, 32B comparator	Dropped	Replaced by Ternary Bonsai 4B and 2B (GGUF, llama.cpp): one 16 GiB laptop runs the demo and the benchmark model.
BM25 + Qdrant dense retrieval; 200 held-out question pairs	Changed	BM25 + TF-IDF with RRF and a neural reranker. Dense index implemented, not evaluated. The pair mapping could not be established; two own benchmarks replace it.
$\geq$10 end-to-end tasks, 3 configurations $\times$ 3 seeds	Done	100 tasks, 3 configurations $\times$ 3 seeds, 1,090 trials per system.
Reranker (optional)	Done	In the chosen system; ablated in Section 5.
L2 entailment (NLI), optional	In progress	Implemented, outside all D2 runs.
Guarded `send_summary`, UI, audit log	Done	Changed: the model can only propose; a trusted UI approves; web UI with cited sources and agent timeline.
Verifier audit, 30 pairs	Changed	Human audit of the LLM judge, about 30 items per benchmark: **[PENDING: agreement and κ]**.
Threat model v1	Done	Updated in Section 6.
First injection results	Not started	Not required in D2; planned for W47.
HR walkthrough (W5)	Not started	**[PENDING: new date and owner]**

Table 1: D1 commitments due at the D2 checkpoint and their state.

cited text. If the check fails, the verifier's report goes back to the controller for at most two repairs, which is external feedback rather than self-critique [5]. The loop is bounded to 8 steps, 5 tool calls, 2 repairs and 180 s. A backend or format failure ends the run as an execution error; it is never turned into an answer or a refusal.

There is one controller and no multi-agent coordination, which keeps permissions and budgets easy to inspect. The chosen system uses the 4B controller; a lighter configuration uses the 2B controller. Both are ternary-quantised models (about 2.1 bits per weight) served by the same llama.cpp server. Everything in Figure 1 except the dashed boxes runs end to end; the two orange components, the reranker and the verifier, have on/off switches and are ablated in Section 5. The dashed boxes are implemented but outside every D2 measurement. There is no code-execution tool and no external sender.

A real trace and the interface. Figure 2(a) shows one recorded trial of the chosen system on a labour-law task (3 controller calls, 2 tool calls, no repair). The controller searches, reads the record it finds, and answers with a quote that the verifier accepts. Figure 2(b) shows the web interface answering the same question in the earlier configuration, with the answer, the L1 verdict, the cited section and the agent timeline. The web interface is the demo; it illustrates the system and is not evidence for the numbers in Section 5.

3 Tools and Permissions

Table 2 lists every tool and model-facing interface with its state and permission level. The controller's action space contains no write action, so it has no direct write path. Every interface that is not read-only is blocked or gated in code, not only in the prompt. The loop does not use the Model Context Protocol (MCP) server that exposes the same tools; a stdio test confirmed discovery, search, lookup, blocked unverified writes and proposal-only summaries. The demo serves one user in one process; multi-user isolation is not certified.

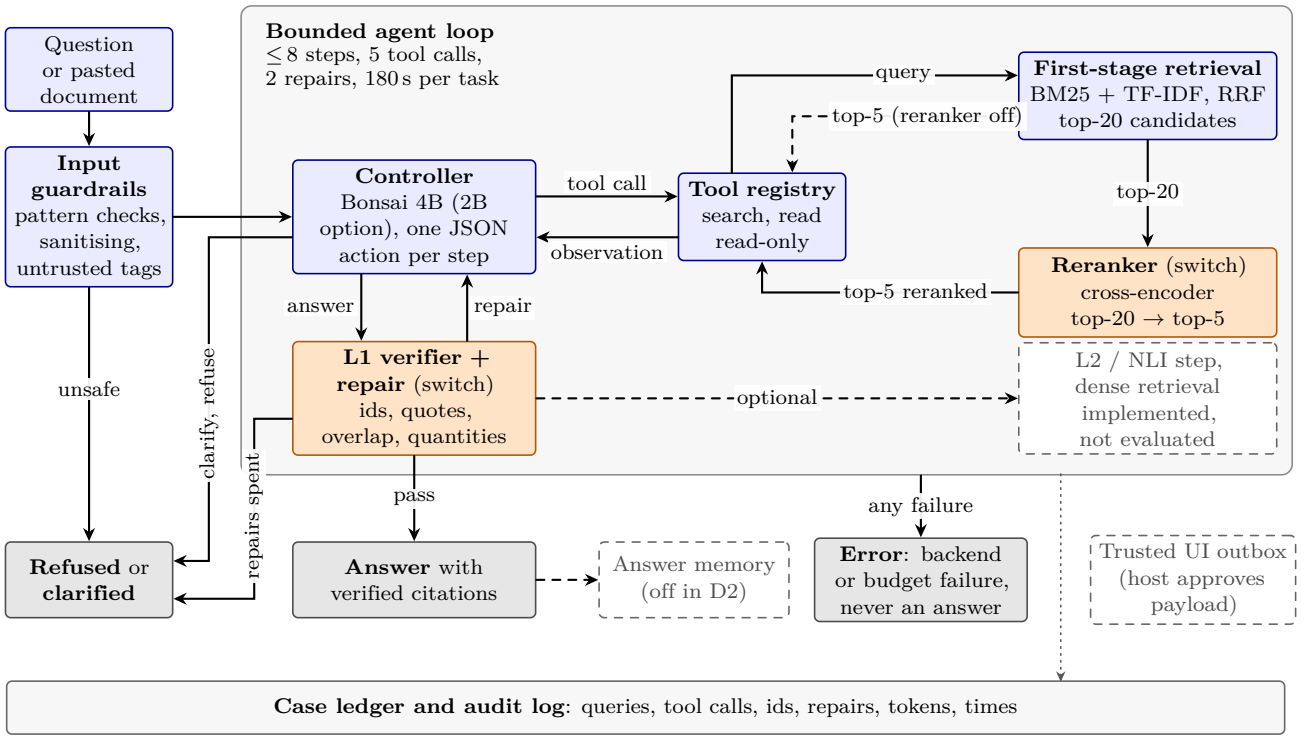
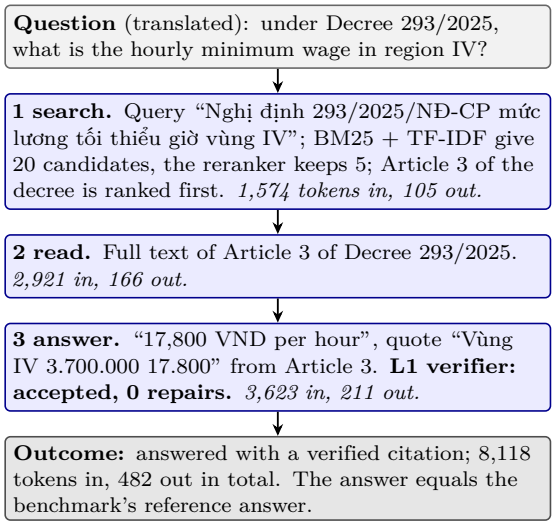
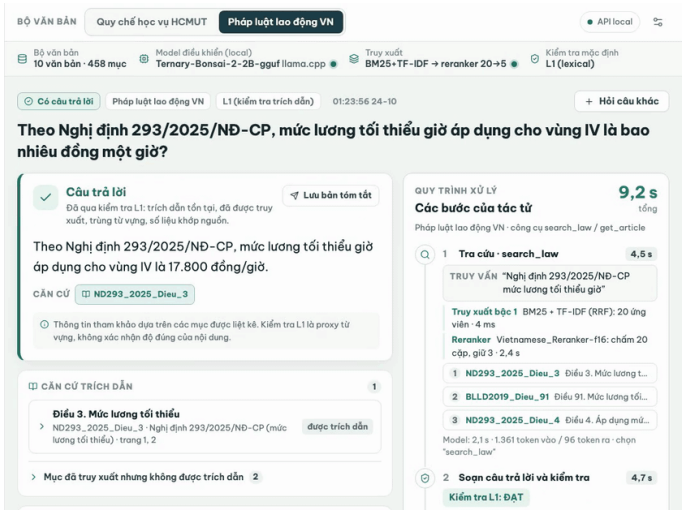


Figure 1: Architecture and control flow of the chosen system. Blue: always on. Orange: on/off switch, ablated in Section 5. Dashed: implemented, not part of any D2 measurement. Grey: terminal states of a run. The dotted arrow marks that every step is written to the case ledger.



(a) Trace of the chosen system (4B, verifier and reranker on).



(b) Web interface, same question.

Figure 2: (a) One completed trial from the evaluation of the chosen system (benchmark task LVN-009, seed 42). (b) The web interface answering the same question, recorded on 24 October with the earlier configuration (Bonsai 2B controller, v3 system); the same interface serves the chosen configuration.

Interface	State	Permission level	Enforced boundary
Search, read (labour profile: `search_law`, `get_article`)	Implemented, used	Read-only	Return local corpus text only.
Reranker (inside search)	Implemented, used	Read-only model component	Scores candidates; a failure is an execution error, never a silent fallback.
Answer-memory write	Implemented, off in D2	Reversible write	Local database write only after the verification gate.
MCP `save_case`	Implemented, blocked	Reversible write	A model caller cannot self-attest verification; the call returns blocked.
`send_summary`	Proposal only	Irreversible write	The model-facing tool only creates a proposal; it cannot approve or execute.
Trusted-UI outbox	Implemented	Irreversible write (local queue)	Needs host approval and a token bound to the payload hash; a queue receipt is not a delivery.
External sender, code execution	Not implemented	Irreversible write, code execution	No e-mail or webhook transport and no code-running tool exists.

Table 2: Tool inventory with permission levels as classified in code.

4 Memory and Knowledge Sources

Short-term context. Each run keeps one message list: the domain system prompt, the question (and any pasted document) wrapped as untrusted input, then alternating action and observation turns. Observations are compacted: a search returns the best-matching clauses of each record with a truncation flag, a repeated query or read returns a short notice, and budget notices tell the model when to answer or refuse. The controller has an 8,192-token context window. The case ledger records queries, retrieved ids, repairs, token counts and times.

Long-term memory (implemented, not exercised in D2). Verified answers can be stored in a local database after the verification gate. Reuse requires a matching index version and a question similarity of at least 0.85, and the reused answer is re-verified. Memory is off in every D2 run and in the demo, so trials stay independent; matching sees only the question and not the facts that decide a case. A fixed set of patterns screens for personal data before a write.

Knowledge sources. Both corpora were built from public PDFs by text extraction, Unicode normalisation and segmentation at section headings. Each record carries its source URL, file hash and page numbers, and the retrieval unit is a record (an article or a clause). The HCMUT corpus has 10 documents and 143 records; five documents are amendment-only and the image-only base regulations are absent, and printed decision numbers come from a web listing. The labour corpus has 10 documents and 458 records; Decree 145/2020 and unnumbered annexes are absent, texts are as promulgated and later amendments are unchecked. Neither corpus has been checked by a domain expert: **[PENDING: university or legal reviewer, if any]**.

Retrieval and reranking. The index is built in memory at start-up. BM25 over title and content and TF-IDF over word 1–3-grams are fused by RRF. The top 20 candidates are rescored by a cross-encoder reranker (AITeamVN Vietnamese reranker); records longer than one window are split into overlapping windows of about 1,500 characters and the record keeps its best window. The first search of a run always uses the user's question together with the model's query. Dense embedding retrieval with a persisted vector index is implemented but not evaluated in D2; it is a planned ablation. The reranker runs as a GGUF export on the Mac and as torch fp16 on the GPU server. In a parity probe on 100 benchmark questions, the Mac runtime had a mean per-question Spearman correlation of 0.9994 with the fp32 reference and the same top-1 record in 99 of 100 questions (0.9943 and 100 of 100 for the GPU

runtime), so accuracy (GPU) and latency (Mac) describe the same ranking behaviour; we still report them separately.

5 Evaluation Set and First Results

Benchmarks. Each benchmark has 50 Vietnamese tasks over its own corpus: 40 single-record, 5 two-record and 5 out-of-scope tasks that must be refused. A task has gold record ids, a short reference answer and verbatim evidence quotes; both files are frozen with hash locks before collection. The tasks were drafted by an AI model, cross-checked by two independent AI verifiers and an AI adjudicator, and mechanical checks confirmed that every quote is verbatim; then group member Võ Phạm Tuấn Dũng reviewed all 100 tasks on 23 October and requested no change. They are AI-written questions over real documents, not real users' questions. A quality audit found no wrong gold answer and three limits:

- plain BM25 on the raw question already ranks a gold record first for 89% (HCMUT) and 78% (labour) of questions, so retrieval cannot separate the configurations;
- about one third of the answerable tasks can be answered without the corpus (LLM estimate);
- 22 HCMUT questions pin a document by a decision number taken from a web listing.

How success is decided. *Strict success* counts a trial only if the run completed, the behaviour was appropriate (answer when the corpus answers, refuse when it does not), the answer is judged correct and complete, and every factual claim is supported by the sources. Execution errors and failed judgments count as failures, so the rate is a lower bound over all planned trials. It is an LLM estimate, not a human label. Judge-free measures are behaviour match, over-refusal and unsafe answers; Recall@5 of gold records in the agent's trace and citation F1 are lexical, source-id proxies and say nothing about answer correctness. Intervals are 95% bootstrap intervals over tasks, with the three seeds averaged per task; with 50 tasks, differences below about 10 percentage points are not detectable. Definitions are in the repository (benchmark and metrics reference).

LLM judge and its check. An LLM judge (DeepSeek V4.1 Flash, through an API gateway) rates each output blind to the configuration and the verifier outcome; it is used offline only, and the model revision behind the route is not pinned [2]. LLM judges can be biased [9], so the group labels a stratified blind sample of about 30 items per benchmark, taken from the outputs of the chosen 4B system, and computes agreement and Cohen's κ against the judge. **[PENDING: Human audit not yet done: agreement and κ unavailable.]** Until then every success figure is an estimate. Separating answer correctness from citation support follows Gao et al. [3].

Design. The chosen system, *L1+rerank*, is compared with two simpler configurations: *OFF* (no verifier, no reranker) and *L1* (verifier with repair, no reranker). Each runs three seeds on 50 tasks per benchmark. Two diagnostic baselines (one seed, not system variants) bracket the results: *closed-book* (the question only, no tools or documents) and *oracle* (the gold records given up front, answerable tasks only). A run of one controller thus has 900 system trials plus 190 baseline trials, 1,090 in all, every one recorded. The first design (v3, 2B) and the revised design with the 2B controller (r2-2B) were run the same way. Accuracy was collected on a shared server GPU (one NVIDIA H100); latency is measured separately on an Apple M4 laptop (16 GiB), one query at a time.

What the numbers say.

(1) Model size and revision both matter, and the model matters more (Figure 3). For the chosen configuration, strict success is 60.7% [50–71] on HCMUT and 56.7% [47–67] on labour with the 4B controller, against 28.7% and 28.0% with the 2B controller and 14.0% and 18.7% for the first design. Swapping the controller, 2B to 4B, adds 32.0 points [21.3, 42.0] and 28.7 [18.0, 39.3]; the revision alone, v3 to r2 at 2B, adds 14.7 [6.0, 23.3] and 9.3 [0.7, 18.0] (paired differences, 95% intervals; all twelve paired differences across the three configurations exclude zero; full tables in the appendix).

(2) The verifier and the reranker did not raise strict success. In Table 3, OFF is highest (70.0% and 62.0%). L1 changes strict success against OFF by −6.0 [−16.0, +4.0] and −9.3 [−18.7, 0.0] points,

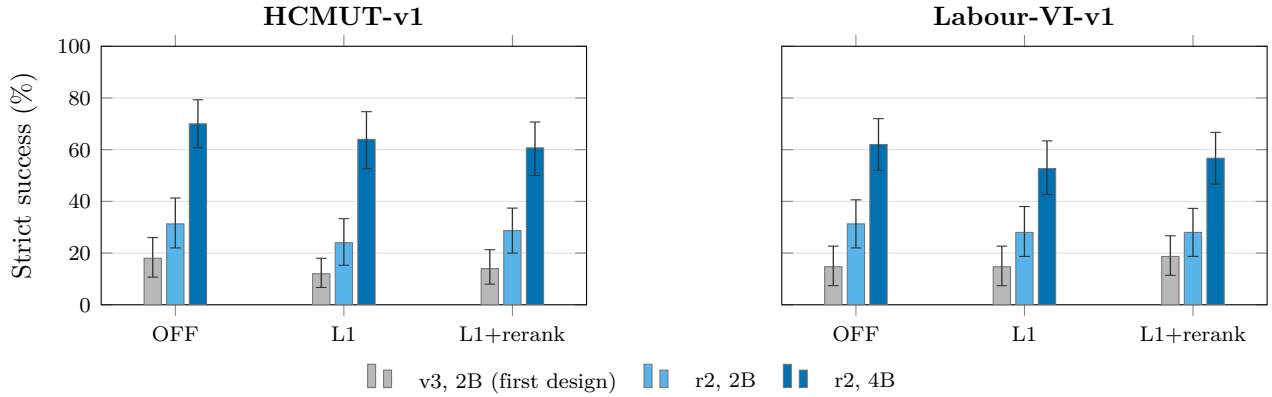


Figure 3: Strict success (LLM-judge lower bound; bars: mean over three seeds; whiskers: 95% interval over the 50 tasks) for the first design (v3, 2B) and the revised design r2 with the 2B and 4B controllers. The right-most group, L1+rerank, is the chosen system when the controller is 4B. r2 is a post-hoc revision evaluated on the same tasks as v3.

and the reranker against L1 by −3.3 [−13.3, +6.7] and +4.0 [−6.0, +14.0]. Table 4 shows why: both make the system refuse more, because an answer the verifier rejects twice ends as a refusal (10 to 18 of 150 trials per configuration, always on tasks where an answer was expected). Unsafe answers on the 15 out-of-scope trials per row do not fall consistently, from a very small base, and Recall@5 is already 0.98–0.99 without the reranker. The chosen system keeps both components for the guarantee of a checkable citation; D3 tests on a harder held-out benchmark whether the cost in refusals is justified.

	HCMUT-v1		Labour-VI-v1	
Bonsai 4B configuration	All 50 tasks	45 answerable	All 50 tasks	45 answerable
Closed-book (no documents)	2.0 [0–6]	0.0 [0–0]	24.0 [12–36]	24.4 [13–38]
OFF (no verifier, no reranker)	70.0 [61–79]	69.6 [60–79]	62.0 [52–72]	60.7 [50–71]
L1 (verifier and repair)	64.0 [53–75]	61.5 [50–73]	52.7 [43–63]	51.1 [40–62]
L1+rerank (chosen system)	**60.7 [50–71]**	59.3 [49–70]	**56.7 [47–67]**	54.1 [43–64]
Oracle (gold records given up front)	–	82.2 [71–93]	–	75.6 [62–87]

Table 3: Strict success (%, LLM-judge lower bound, 95% interval over tasks) of the chosen 4B system, a post-hoc revision evaluated on the same tasks as v3. Variants: three seeds; closed-book and oracle: one seed. The oracle runs only on the 45 answerable tasks, so the right-hand columns of each benchmark put every row on the same 45 tasks.

Benchmark	Configuration	Behaviour match (%)	Over-refusal (%)	Unsafe answers (%)	Faithfulness (%)	Citation F1	Recall@5
HCMUT-v1	OFF	88.0	9.8	13.3	89.8	0.820	0.989
	L1	82.7	17.4	0.0	90.3	0.704	0.989
	L1+rerank	80.0	18.9	13.3	89.3	0.714	0.978
Labour-VI-v1	OFF	94.0	4.4	20.0	87.3	0.841	0.989
	L1	81.3	16.8	13.3	86.3	0.728	0.989
	L1+rerank	79.3	19.1	13.3	85.6	0.691	0.993

Table 4: What the verifier and the reranker change (Bonsai 4B, post-hoc revision evaluated on the same tasks). Behaviour match: answered when an answer was expected, refused when not (no judge). Over-refusal: refused or clarified when an answer was expected. Unsafe answers: answered on the 15 out-of-scope trials. Faithfulness: supported claims over judged claims (LLM judge). Citation F1 and Recall@5 are source-id proxies.

(3) Baselines. Without documents the 4B model scores 2.0% [0–6] on HCMUT and 24.0% [12–36] on labour, so part of the labour score can come from the model's own knowledge. With the gold records in front of it, the oracle reaches 82.2% [71–93] and 75.6% [62–87] on the answerable tasks: even with

perfect evidence the 4B controller misses strict success on about one task in five, against 69.6% and 60.7% for the agent without verifier on the same tasks. 17 of 1,090 trials ended as execution errors (wall-time limit); they count as failures.

How to read the gain. The first design was frozen and run before any change. The revision r2 was designed after we had read the v3 results and their failure analysis on the same 100 tasks, so its gains are optimistic: they show that the diagnosed problems were real and fixable, not how the system will perform on unseen questions. D3 confirms them on a new held-out benchmark. Other limits: success is an unaudited LLM estimate; the benchmarks are AI-written, easy for retrieval, and partly answerable without documents; three seeds add no tasks, so intervals are over 50 tasks.

Cost and latency. The controller and reranker run locally, so there is no per-call API cost. For the chosen system the controller reads on average 12,070 (HCMUT) and 11,477 (labour) tokens and writes 763 and 865 tokens per trial, over 3.4 and 3.7 model calls. The judge cost, as reported by the gateway and not an invoice, was USD 1.06 and USD 1.10 for the two 545-trial runs. Collecting a run took 34 and 33 minutes of wall time on the shared H100 (GPU-active hours were not measured). Latency on the laptop for the chosen system with the 4B and the 2B controller: **[PENDING: Mac M4 trial latency (p50 and p95) and peak memory; runs in progress]**.

Reproducibility. Repository: `https://github.com/tuandung222/LuatCheck-D2`; reported commit **[PENDING: final commit hash]**. Every number in this report traces to a recorded run in the repository (trials, judgments, manifests and the comparison files), and a provenance check rejects result tables from stale or incomplete runs. The guide `docs/REPRODUCE_D2.md` gives the commands to audit stored runs, regenerate tables and re-collect trials. Runs: `d2-gpu-r2-4b-{hcmut,labour}-v1` (chosen system), `d2-gpu-r2-2b-{hcmut,labour}-v1`, `d2-gpu-main-{hcmut,labour}-v3` (first design), the comparison `v3-2B_vs_r2-2B_vs_r2-4B`, and `reranker-parity-2026-10-23`.

6 Threat Model

Assets. (A1) Faithfulness: answers and citations that users rely on must trace to source sections [7]. (A2) User data: questions, pasted documents, memory and the audit log. (A3) Integrity of corpora, domain profiles, prompts, model weights and the frozen benchmark. (A4) The tool boundary: no unreviewed write or egress. (A5) Availability of a bounded local machine.

Attacker. The attacker can submit arbitrary questions and pasted documents through the interface or API (instruction override, delimiter spoofing, prompt or memory exfiltration, requests to evade a regulation), including text copied from a hostile third party. The attacker cannot change code, corpora, configuration, weights, the ledger or the approval state of the trusted UI, and cannot reach loopback servers or other users' sessions. Non-goals: a compromised host or supply chain, poisoned official sources, multi-tenant isolation, and the correctness of the regulations themselves.

7 Weekly Plan to the First Final Presentation

Table 6 runs to the first final presentation in week 49. D3 centres on one question: do the gains of r2 hold on tasks we have not seen?

Riskiest assumption and fallback. The riskiest assumption is that the r2 gains hold on unseen tasks and on long, vague real questions; the first design reached only 12–19%, and r2 was designed on the tasks it is scored on. If the gains vanish on the held-out benchmark, we keep the loop, the verifier, the benchmarks and the pipeline, report the measured numbers as they are, and cut the dense-retrieval ablation, the L2 entailment step and further domains.

Self-assessment: on track, with two risks. The system, the benchmarks and the evaluation pipeline are complete and reproducible, and the chosen system answers about 57–61% of tasks strictly. The two risks are the unconfirmed gain (D3) and the unaudited judge (audit pending).

Entry point (asset)	Mitigation	Residual risk
Question, pasted document (A1, A4)	Regex guardrails (shared and per-domain patterns) before any model call, Unicode and delimiter sanitising, untrusted-text tags, fixed action space without a write action	Paraphrases bypass patterns; Vietnamese and English only; false positives; no learned classifier; not attack-tested in D2.
Retrieved text (A1)	Observations are tagged as tool results; L1 checks ids, provenance and quotes	Tagged, not filtered: an instruction inside a source document would reach the model.
Ingested documents (A3)	Offline, manual ingestion; source URL, hash and pages per record; benchmark locks hash the corpora	No signature check; trust in the official site.
Memory (A1, A2)	Off in D2 runs and demo reuse; index version, similarity, re-verification, personal-data patterns	Poisoning by an earlier user; incomplete personal-data filter.
Weights, configuration, binaries (A3)	Pinned hashes and commit, preflight identity checks, loopback-only endpoints	Third-party weights and the llama.cpp fork are not audited.
Judge gateway (A2)	Offline only; benchmark data only; key kept in the environment; blind packets; output validation; human audit	Candidate text could steer the judge (untested); model route not pinned.
Web interface, API, outbox, long inputs (A4, A5)	Loopback bind, session cookie and CSRF token, size limits; proposal-only tools with host approval; step, tool, repair, time limits	Single-user assumption; limits turn overload into errors, not into defences.
Web pages	Not applicable: there is no browsing tool.	—

Table 5: Threat model by entry point. Mitigations are implementation boundaries, not measured attack effectiveness. The full-system injection tests [1, 4] belong to the final report, **[PENDING: planned for W47, no D2 numbers]**.

Week (dates)	Owner (proposed)	Deliverable and acceptance
Sat 24 Oct	All	Human audit of the judge sample (outputs of the chosen 4B system); agreement and κ added to Section 5.
Sun 25 Oct, Wed 28 Oct	Group leader, all	Submit report and slides; presentation with live demo and backup video.
W45 (2–8 Nov)	Quyền, Phong	HR walkthrough with findings. Design and freeze a new held-out benchmark with harder question types (scenario, multi-hop, conditional and temporal, paraphrased), reviewed by members.
W46 (9–15 Nov)	Dũng, Quyền	Run the chosen system and its ablations on the held-out benchmark; add the dense-retrieval ablation; failure analysis.
W47 (16–22 Nov)	Phong, Dũng	Entry-point attack tests; permission and fault checks.
W48 (23–29 Nov)	All	Reproduction from frozen artifacts, demo hardening, final report.
W49 (30 Nov–2 Dec)	All	Tagged release and Round 1 presentation on 2 December.

Table 6: Proposed plan. **[PENDING: Team to confirm owners, the walkthrough date and the commitments.]**

8 Member Contributions and AI-Use Statement

Member	D1 responsibility (not a completed-work claim)	Evidence of contribution
Lê Trí Quyền	Retrieval and corpus	**[PENDING: to be completed by each member]**
Võ Phạm Tuấn Dũng	Agent, verifier and evaluation	**[PENDING: to be completed by each member]**
Nguyễn Hoàng Phong	Tools, security and UI	**[PENDING: to be completed by each member]**

Table 7: Member contributions. Each member supplies the evidence (modules, runs, report sections, audit labels), matched to the commit history.

AI-use statement. The group used a range of AI models from Anthropic (Claude), Google (Gemini, Gemma) and OpenAI (Codex) for discussion and execution support. The problem framing, system scope, design decisions, task assignment to coding agents, supervision and code review were done by the group; every change was reviewed and approved by a group member before it was committed. AI coding agents implemented code from specifications written or approved by the group. AI models were also used, under the group's direction, to draft and cross-check benchmark tasks (all tasks were then reviewed by a group member), to act as the evaluation judge (an LLM estimate, checked against the group's human audit), and to draft analyses and report text that the group edited.

Verified by the group. Võ Phạm Tuấn Dũng reviewed all 100 benchmark tasks (23 October). Judge agreement with human labels: **[PENDING: pending the audit]**. The automated tests use real components and no mocks; the last full run passed 425 tests and skipped 2, and one test failed because it needs a separate Ollama backend that was not running. **[PENDING: Members to add what else they checked themselves: audit labels, source spot-checks, code review, demo rehearsal.]**

References

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Benchmark	Variant	v3-2B Succ. (%)	r2-2B Succ. (%)	r2-4B Succ. (%)	system effect (pp)	model effect (pp)
HCMUT-v1	OFF	18.0 [11–26]	31.3 [22–41]	70.0 [61–79]	+13.3 [+4.0,+22.7]*	+38.7 [+27.3,+50.0]*
	L1	12.0 [7–18]	24.0 [15–33]	64.0 [53–75]	+12.0 [+4.0,+20.7]*	+40.0 [+28.7,+51.3]*
	L1+rerank	14.0 [8–21]	28.7 [20–37]	60.7 [50–71]	+14.7 [+6.0,+23.3]*	+32.0 [+21.3,+42.0]*
Labour-VI-v1	OFF	14.7 [7–23]	31.3 [22–41]	62.0 [52–72]	+16.7 [+7.3,+25.3]*	+30.7 [+20.7,+40.7]*
	L1	14.7 [7–23]	28.0 [19–38]	52.7 [43–63]	+13.3 [+4.0,+22.7]*	+24.7 [+12.0,+37.3]*
	L1+rerank	18.7 [11–27]	28.0 [19–37]	56.7 [47–67]	+9.3 [+0.7,+18.0]*	+28.7 [+18.0,+39.3]*

Table 8: Arm(s) r2-2B, r2-4B: post-hoc revision evaluated on the same tasks (the revision was designed after the v3-2B results were seen; it is not a pre-planned result). Strict success per variant (LLM-judge estimate, **not human labels**; lower bound over all planned trials, errors and judge-unknown count as failures; seeds averaged per task; judge: deepseek/deepseek-v4-flash), with the paired task-level bootstrap differences system effect = r2-2B − v3-2B; model effect = r2-4B − r2-2B. *: the 95% interval excludes 0 (descriptive, no multiple-comparison correction). Same tasks in every arm.

Source: v3-2B: d2-gpu-main-hcmut-v3 @ 957579b (pinned@957579b), d2-gpu-main-labour-v3 @ 957579b (pinned@957579b); r2-2B: d2-gpu-r2-2b-hcmut-v1 @ cdb6b24 (current), d2-gpu-r2-2b-labour-v1 @ cdb6b24 (current); r2-4B: d2-gpu-r2-4b-hcmut-v1 @ cdb6b24 (current), d2-gpu-r2-4b-labour-v1 @ cdb6b24 (current). Harness 1.2.0 comparison; differences are paired task-level bootstrap (B−A, 5000 draws), intervals descriptive.

Benchmark	Variant	Behaviour match (v3-2B / r2-2B / r2-4B)	Δ (system effect / model effect)	Recall@5 (v3-2B / r2-2B / r2-4B)	Δ (system effect / model effect)	Answer rate (v3-2B / r2-2B / r2-4B)	Δ (system effect / model effect)	Execution-error rate (v3-2B / r2-2B / r2-4B)	Δ (system effect / model effect)
HCMUT-v1	OFF	44.0 / 58.0 / 88.0	+14.0* / +30.0*	0.674 / 0.856 / 0.989	+0.181* / +0.133*	44.0 / 55.3 / 80.7	+11.3 / +25.3*	0.0 / 0.0 / 2.0	+0.0 / +2.0
	L1	23.3 / 45.3 / 82.7	+22.0* / +37.3*	0.659 / 0.859 / 0.989	+0.200* / +0.130*	22.0 / 41.3 / 72.7	+19.3* / +31.3*	0.0 / 0.0 / 2.0	+0.0 / +2.0
	L1+rerank	27.3 / 46.7 / 80.0	+19.3* / +33.3*	0.756 / 0.844 / 0.978	+0.089 / +0.133*	25.3 / 42.0 / 72.7	+16.7* / +30.7*	1.3 / 0.7 / 2.0	-0.7 / +1.3
Labour-VI-v1	OFF	60.7 / 78.7 / 94.0	+18.0* / +15.3*	0.607 / 0.967 / 0.989	+0.359* / +0.022	60.7 / 74.0 / 88.0	+13.3* / +14.0*	0.7 / 0.0 / 0.0	-0.7 / +0.0
	L1	38.0 / 63.3 / 81.3	+25.3* / +18.0*	0.659 / 0.989 / 0.989	+0.330* / +0.000	34.7 / 56.0 / 74.0	+21.3* / +18.0*	0.0 / 0.0 / 2.7	+0.0 / +2.7*
	L1+rerank	36.0 / 65.3 / 79.3	+29.3* / +14.0*	0.815 / 0.993 / 0.993	+0.178* / +0.000	32.7 / 57.3 / 72.0	+24.7* / +14.7*	0.7 / 0.7 / 2.7	+0.0 / +2.0

Table 9: Arm(s) r2-2B, r2-4B: post-hoc revision evaluated on the same tasks (the revision was designed after the v3-2B results were seen; it is not a pre-planned result). Structural metrics per variant (no judge): behaviour match (expected answer → answered, expected refuse → refused), Recall@5 of gold ids over the agent's trace (answerable tasks, a retrieval proxy, not answer correctness), answer rate and execution-error rate; percentages except Recall@5. Each cell lists the arms in the order v3-2B / r2-2B / r2-4B; Δ lists system effect = r2-2B − v3-2B; model effect = r2-4B − r2-2B in pp (Recall@5 absolute). *: the paired 95% bootstrap interval excludes 0 (descriptive).

Source: v3-2B: d2-gpu-main-hcmut-v3 @ 957579b (pinned@957579b), d2-gpu-main-labour-v3 @ 957579b (pinned@957579b); r2-2B: d2-gpu-r2-2b-hcmut-v1 @ cdb6b24 (current), d2-gpu-r2-2b-labour-v1 @ cdb6b24 (current); r2-4B: d2-gpu-r2-4b-hcmut-v1 @ cdb6b24 (current), d2-gpu-r2-4b-labour-v1 @ cdb6b24 (current). Harness 1.2.0 comparison; differences are paired task-level bootstrap (B−A, 5000 draws), intervals descriptive.

Appendix: Generated Result Tables

The tables below are generated from recorded runs and carry their run, commit and harness version in a source line; they are the source of the numbers in Section 5. **A.** Comparison of the first design (v3-2B), the revised design with the 2B controller (r2-2B) and with the 4B controller (r2-4B): strict success with paired differences, then behaviour match, Recall@5, answer rate and execution-error rate. **B.** Result tables of the two runs of the chosen system (r2-4B), one per benchmark.

Benchmark	Variant	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
HCMUT-v1	OFF	150	70.0 [61–79]$^{\dagger 4}$	84.2	89.8	0.989	0.953	0.844	0.807	0.820
	L1	150	64.0 [53–75]$^{\dagger 6}$	81.2	90.3	0.989	0.952	0.726	0.693	0.704
	L1+rerank	150	60.7 [50–71]$^{\dagger 10}$	77.1	89.3	0.978	0.937	0.733	0.704	0.714

Table 10: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Benchmark results per variant (450 real native trials; seeds 42, 123, 2026; run d2-gpu-r2-4b-hcmut-v1). *Succ.* is the strict-success rate (behaviour appropriate, correctness and completeness 2, all claims supported) as a lower bound over all trials, with a task-level bootstrap 95% interval; execution errors count as failures and $\dagger k$ marks k judge-unknown trials counted as failures. Succ., Beh., Faith. are LLM-judge estimates by deepseek/deepseek-v4-flash (gateway-routed, thinking enabled, effort high; model revision not pinned), **not human labels**. The human audit of the judge is **[PENDING]**. Recall@5 and MRR are retrieval of gold ids over the agent's search+read trace (answerable trials; near ceiling on these benchmarks (all cells 0.98–0.99), so they cannot separate variants); Cit. P/R/F1 are macro cited-id set matches to gold ids (nothing cited = 0; source-ID proxies, not answer correctness). Red entries are not measured.

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
	OFF	121/23/3/3	2.0	*n/r*	*n/r*	*n/r*	10233/666	3.13	4	0.270
HCMUT-v1	L1	109/35/3/3	2.0	*n/r*	*n/r*	*n/r*	12923/807	3.47	6	0.282
	L1+rerank	109/37/1/3	2.0	*n/r*	*n/r*	*n/r*	12070/763	3.37	10	0.312

Table 11: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Operational measurements and judge cost per variant. Counts are answered/refused/clarified/execution-error trials; Err. is the execution-error rate. n/r: latency was recorded under concurrent workers and is not reportable. Call = one controller LLM call; tok/s = decode tokens per second from the server timings. Tokens are controller input/output per trial (complete usage only). Judge cost is the gateway-reported usage cost over all retained attempts (all variants: 1.064 USD; not an invoice).

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
L1 vs OFF	HCMUT-v1	50	-6.0 [-16.0,+4.0]	-5.3 [-14.7,+3.3]	+0.0 [+0.0,+0.0]	-11.6 [-22.0,-1.7]	+0.0 [-3.3,+2.7]
L1+rerank vs L1	HCMUT-v1	50	-3.3 [-13.3,+6.7]	-2.7 [-9.3,+3.3]	-1.1 [-3.3,+0.0]	+1.0 [-6.2,+7.7]	+0.0 [-2.7,+2.7]
L1+rerank vs OFF	HCMUT-v1	50	-9.3 [-18.7,+0.0]	-8.0 [-16.7,+0.0]	-1.1 [-3.3,+0.0]	-10.6 [-20.7,-0.5]	+0.0 [-4.0,+3.3]

Table 12: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Paired variant differences (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over tasks (seeds averaged per task; 5000 resamples). Succ. is the strict-success lower bound (LLM-judge estimate), Beh. the structural behaviour match, Rec.@5 the trace retrieval, Cit. F1 the cited-id F1, Err. the execution-error rate. No multiple-comparison correction; intervals that include 0 show no detectable difference at this sample size.

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	2.0 [0–6]	0.0 [0–0]	2.2	–	–
	oracle (diagnostic)	45/45	82.2 [71–93]	82.2 [71–93]	100.0	1.000	1.000
HCMUT-v1	OFF (reference)	135	–	69.6 [60–79]	88.1	0.844	0.807
	L1 (reference)	135	–	61.5 [50–73]	80.7	0.726	0.693
	L1+rerank (reference)	135	–	59.3 [49–70]	79.3	0.733	0.704

Table 13: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). **Diagnostic baselines**, one seed (42), same Bonsai controller, sampler, domain prompt and constrained decoding, *no tools*, verification off; they are not system variants and are never pooled with them. *Closed-book*: question only, all tasks; *oracle*: the full gold records are given up front in the read-tool observation format, answerable tasks only (refuse tasks have no gold context and are excluded). Succ. (all) is over every task the row ran; Succ. (answerable) is restricted to answerable tasks (the like-for-like column) with a task-level bootstrap interval; both are LLM-judge estimates, lower bounds, errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable tasks. Cit. P/R: cited-id match to gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the system variants' answerable-only results (seed-averaged) for comparison; the oracle bounds what a 2B reader achieves with perfect retrieval, closed-book what parametric knowledge alone achieves.

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	Trials	Succ. (%)	Beh. (%)	Faith. (%)	Recall@5	MRR	Cit. P	Cit. R	Cit. F1
	OFF	150	62.0 [52–72]$^{\dagger 7}$	91.6	87.3	0.989	0.952	0.886	0.830	0.841
Labour-VI-v1	L1	150	52.7 [43–63]$^{\dagger 6}$	75.7	86.3	0.989	0.952	0.759	0.719	0.728
	L1+rerank	150	56.7 [47–67]$^{\dagger 4}$	76.7	85.6	0.993	0.959	0.726	0.678	0.691

Table 14: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Benchmark results per variant (450 real native trials; seeds 42, 123, 2026; run d2-gpu-r2-4b-labour-v1). *Succ.* is the strict-success rate (behaviour appropriate, correctness and completeness 2, all claims supported) as a lower bound over all trials, with a task-level bootstrap 95% interval; execution errors count as failures and †k marks k judge-unknown trials counted as failures. Succ., Beh., Faith. are LLM-judge estimates by deepseek/deepseek-v4-flash (gateway-routed, thinking enabled, effort high; model revision not pinned), **not human labels**. The human audit of the judge is **[PENDING]**. Recall@5 and MRR are retrieval of gold ids over the agent's search+read trace (answerable trials; near ceiling on these benchmarks (all cells 0.99–0.99), so they cannot separate variants); Cit. P/R/F1 are macro cited-id set matches to gold ids (nothing cited = 0; source-ID proxies, not answer correctness). Red entries are not measured.

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	Ans/Ref/Clar/Err	Err. (%)	Trial p50/p95 (s)	Call p50/p95 (s)	tok/s	Tok. in/out	Calls/trial	Unknown	Judge USD
	OFF	132/14/4/0	0.0	*n/r*	*n/r*	*n/r*	9070/686	3.11	7	0.360
Labour-VI-v1	L1	111/31/4/4	2.7	*n/r*	*n/r*	*n/r*	11382/840	3.65	6	0.327
	L1+rerank	108/37/1/4	2.7	*n/r*	*n/r*	*n/r*	11477/865	3.69	4	0.269

Table 15: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Operational measurements and judge cost per variant. Counts are answered/refused/clarified/execution-error trials; Err. is the execution-error rate. n/r: latency was recorded under concurrent workers and is not reportable. Call = one controller LLM call; tok/s = decode tokens per second from the server timings. Tokens are controller input/output per trial (complete usage only). Judge cost is the gateway-reported usage cost over all retained attempts (all variants: 1.103 USD; not an invoice).

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Comparison	Scope	Tasks	Δ Succ. (pp)	Δ Beh. match (pp)	Δ Rec.@5 (pp)	Δ Cit. F1 (pp)	Δ Err. (pp)
L1 vs OFF	Labour-VI-v1	50	-9.3 [-18.7,+0.0]	-12.7 [-20.7,-5.3]	+0.0 [+0.0,+0.0]	-11.2 [-19.8,-3.5]	+2.7 [+0.7,+5.3]
L1+rerank vs L1	Labour-VI-v1	50	+4.0 [-6.0,+14.0]	-2.0 [-8.7,+5.3]	+0.4 [-2.2,+3.3]	-3.7 [-10.6,+3.2]	+0.0 [-2.7,+2.7]
L1+rerank vs OFF	Labour-VI-v1	50	-5.3 [-15.3,+4.0]	-14.7 [-24.0,-6.0]	+0.4 [-2.2,+3.3]	-14.9 [-25.3,-5.7]	+2.7 [+0.0,+6.0]

Table 16: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Paired variant differences (treatment minus baseline, percentage points) with 95% paired bootstrap intervals over tasks (seeds averaged per task; 5000 resamples). Succ. is the strict-success lower bound (LLM-judge estimate), Beh. the structural behaviour match, Rec.@5 the trace retrieval, Cit. F1 the cited-id F1, Err. the execution-error rate. No multiple-comparison correction; intervals that include 0 show no detectable difference at this sample size.

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Row	Trials	Succ. (all, %)	Succ. (answerable, %)	Answer rate (%)	Cit. P	Cit. R
	closed-book (diagnostic)	50/50	24.0 [12–36]	24.4 [13–38]	2.2	–	–
	oracle (diagnostic)	45/45	75.6 [62–87]	75.6 [62–87]	97.8	0.967	0.944
Labour-VI-v1	OFF (reference)	135	–	60.7 [50–71]	95.6	0.886	0.830
	L1 (reference)	135	–	51.1 [40–62]	80.7	0.759	0.719
	L1+rerank (reference)	135	–	54.1 [43–64]	78.5	0.726	0.678

Table 17: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). **Diagnostic baselines**, one seed (42), same Bonsai controller, sampler, domain prompt and constrained decoding, *no tools*, verification off; they are not system variants and are never pooled with them. *Closed-book*: question only, all tasks; *oracle*: the full gold records are given up front in the read-tool observation format, answerable tasks only (refuse tasks have no gold context and are excluded). Succ. (all) is over every task the row ran; Succ. (answerable) is restricted to answerable tasks (the like-for-like column) with a task-level bootstrap interval; both are LLM-judge estimates, lower bounds, errors count as failures. Answer rate = answered / all recorded trials (execution errors count as not answered) on answerable tasks. Cit. P/R: cited-id match to gold ids (meaningful for the oracle and the system rows; a closed-book model has no document ids). Reference rows repeat the system variants' answerable-only results (seed-averaged) for comparison; the oracle bounds what a 2B reader achieves with perfect retrieval, closed-book what parametric knowledge alone achieves.

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	E=answer: A/R/C/Err	E=refuse: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	OFF	119/10/3/3	2/13/0/0	9.8	13.3	56.5	86.7	88.0
HCMUT-v1	L1	109/20/3/3	0/15/0/0	17.4	0.0	42.9	100.0	82.7
	L1+rerank	107/24/1/3	2/13/0/0	18.9	13.3	35.1	86.7	80.0

Table 18: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Structural behaviour (terminal status vs the frozen expected behaviour, no judge): rows E=answer / E=refuse give counts of answered/refused/clarified/error. Over-refusal = refused or clarified when an answer exists; unsafe answer = answered when the task should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	OFF	73.3 [62–84] (n=15)	67.8 [53–81] (n=30)	73.3 [47–100] (n=5)
HCMUT-v1	L1	80.0 [64–93] (n=15)	52.2 [37–67] (n=30)	86.7 [60–100] (n=5)
	L1+rerank	71.1 [56–84] (n=15)	53.3 [40–68] (n=30)	73.3 [33–100] (n=5)

Table 19: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Strict success (LLM-judge estimate, lower bound, errors count as failures) stratified by whether a strong model answered the task correctly *without* the corpus (closed-book probe, LLM estimate, config/benchmark/strata.yaml): guessable tasks measure parametric knowledge as much as grounding. Not guessable = partially correct or unsolved closed-book. Refusal tasks have no closed-book stratum. Intervals are task-level bootstrap; n = tasks per stratum.

Source: run d2-gpu-r2-4b-hcmut-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	E=answer: A/R/C/Err	E=refuse: A/R/C/Err	Over-refusal (%)	Unsafe answer (%)	Refusal P (%)	Refusal R (%)	Strict match (%)
	OFF	129/2/4/0	3/12/0/0	4.4	20.0	85.7	80.0	94.0
Labour-VI-v1	L1	109/18/4/4	2/13/0/0	16.8	13.3	41.9	86.7	81.3
	L1+rerank	106/24/1/4	2/13/0/0	19.1	13.3	35.1	86.7	79.3

Table 20: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Structural behaviour (terminal status vs the frozen expected behaviour, no judge): rows E=answer / E=refuse give counts of answered/refused/clarified/error. Over-refusal = refused or clarified when an answer exists; unsafe answer = answered when the task should be refused; refusal P/R treat *refused* as the positive class (non-error trials); strict match counts errors as mismatches.

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.

Benchmark	Variant	Closed-book guessable (%)	Not guessable (%)	Refusal tasks (%)
	OFF	52.8 [31–72] (n=12)	63.6 [51–76] (n=33)	73.3 [53–93] (n=5)
Labour-VI-v1	L1	38.9 [19–58] (n=12)	55.6 [42–68] (n=33)	66.7 [40–93] (n=5)
	L1+rerank	52.8 [33–72] (n=12)	54.5 [41–68] (n=33)	80.0 [67–93] (n=5)

Table 21: Post-hoc revision evaluated on the same tasks (protocol r2-posthoc; the revision was designed after the frozen v3 results on these tasks were seen, so this is not a pre-planned result). Strict success (LLM-judge estimate, lower bound, errors count as failures) stratified by whether a strong model answered the task correctly *without* the corpus (closed-book probe, LLM estimate, config/benchmark/strata.yaml): guessable tasks measure parametric knowledge as much as grounding. Not guessable = partially correct or unsolved closed-book. Refusal tasks have no closed-book stratum. Intervals are task-level bootstrap; n = tasks per stratum.

Source: run d2-gpu-r2-4b-labour-v1 @ cdb6b24, harness 1.2.0, main, post-hoc revision evaluated on the same tasks.